使用Files操作文件与文件夹

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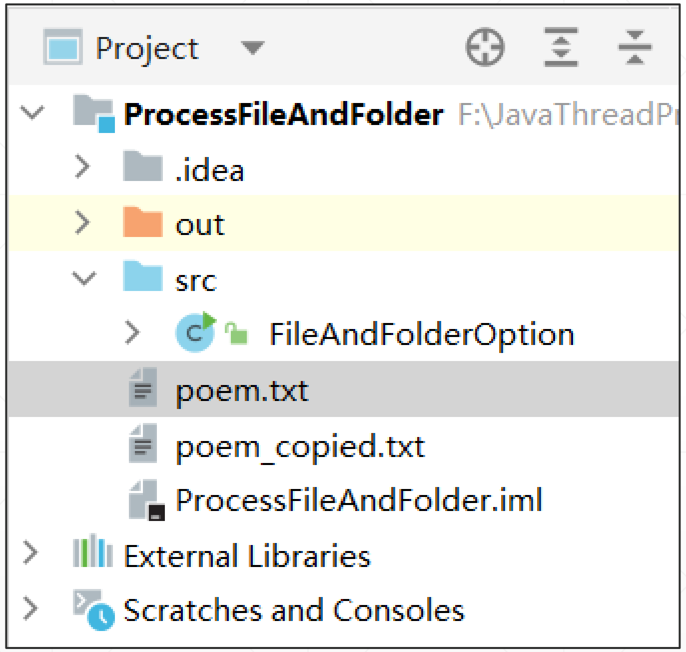
使用Files操作文件

- Java NIO提供了一个Files类，集成了N多实用的文件与文件夹操作功能，可供调用。
- Files类通常与Path类相互配合，完成各种实用文件操作功能。
- 下面我们通过多个示例，展示它的基本用法。

复制文件




```java
//文件复制
private static void copyFile() throws IOException {
    Path source = Paths.get( first: "poem.txt");
    Path target = Paths.get( first: "poem_copied.txt");
    Files.copy(source, target, StandardCopyOption.REPLACE_EXISTING);
    System.out.println("复制结束");
}
```

FileAndFolderOption.java

创建和删除文件

```
//创建和删除文件
private static void createAndDeleteFile() throws IOException {
    Path createdFilePath = Files.createFile(Paths.get( first: "myFile.txt"));
    System.out.println(createdFilePath + " 文件已创建。");
    boolean deleteIfExists = Files.deleteIfExists(Paths.get( first: "myFile.txt"));
    System.out.println("删除结果 = " + deleteIfExists);
}
```

创建临时文件

NEW

```java
//创建临时文件
private static void createTempFile() throws IOException {
    Path tempFile = Files.createTempFile( prefix: "myapp", suffix: ".tmp");
    //C:\Users\JinXu\AppData\Local\Temp\myapp14342790709253233910.tmp
    System.out.println(tempFile);
    //删除它
    Files.deleteIfExists(tempFile);
}
```

创建文件夹

```java
//创建文件夹
private static void createFolder() throws IOException {
    Path newDir = Paths.get( first: "data");
    if (!Files.exists(newDir)) {
        Files.createDirectory(newDir);
        System.out.println("文件夹创建成功");
    } else {
        System.out.println("文件夹已经创建过了。");
    }
}
```

移动文件

```java
//移动文件
private static void moveFile() throws IOException {
    Path sourceFile = Paths.get( first: "poem.txt");
    Path targetDir = Paths.get( first: "./data/poem-moved.txt");
    try {
        Files.move(sourceFile, targetDir, StandardCopyOption.REPLACE_EXISTING);
        System.out.println("文件移动结束");
    } catch (IOException ex) {
        ex.printStackTrace();
    }
}
```

获取文件或文件夹的相关信息

```java
public static void main(String[] args) throws IOException {
    Scanner input = new Scanner(System.in);
    System.out.println("输入文件或文件夹名:");
    Path path = Paths.get(input.nextLine());
    //仅处理真实存在的文件夹
    if (Files.exists(path)) {
        System.out.printf("%n%s exists%n", path.getFileName());
        System.out.printf("%s a directory%n",
                Files.isDirectory(path) ? "Is" : "Is not");
        System.out.printf("%s an absolute path%n",
                path.isAbsolute() ? "Is" : "Is not");
        System.out.printf("Last modified: %s%n", Files.getLastModifiedTime(path));
        System.out.printf("Size: %s%n", Files.size(path));
        System.out.printf("Path: %s%n", path);
        System.out.printf("Absolute path: %s%n", path.toAbsolutePath());
        if (Files.isDirectory(path)) {
            System.out.printf("%nDirectory contents:%n");
            try(DirectoryStream<Path> directoryStream = Files.newDirectoryStream(path)){
                for (Path p : directoryStream)
                    System.out.println(p);
            }
        }
    } else {
        System.out.printf("%s does not exist%n", path);
    }
}
```

GetFileAndDirectoryInfo.java

Run: GetFileAndDirectoryInfo

```
"C:\Program Files\Java\jdk-17\bin\java.exe"
输入文件或文件夹名:
c:\windows

windows exists
Is a directory
Is an absolute path
Last modified: 2021-10-22T23:53:29.0749942Z
Size: 20480
Path: c:\windows
Absolute path: c:\windows
```